SYSDESIGN REFERENCE

SQL vs NoSQL

Core concepts, reference diagram, tradeoffs, and interview prompts.

Select a data model and consistency model from access patterns, not product labels.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / SQL-VS-NOSQL

SYSDESIGN

SQL vs NoSQL

CORE IDEA

Select a data model and consistency model from access patterns, not product labels.

Reference architecture decision chain

01	02	03	04
Transactions	Joins	Schema Evolution	Scale Shape

Decisions to defend

+ Transactions
+ joins
+ schema evolution
+ scale shape
+ indexing
+ and operational maturity.

Failure modes to test

+ Polyglot sprawl
+ denormalization bugs
+ and choosing for hypothetical scale.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

SQL vs NoSQL

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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